

STAA57-Winter 2023: Midterm Test Solution

5pm-7pm; 11 Mar 2023

General Info:

- Total time allowed = 2 hours
 - writing time = 5.00pm - 6.40pm = 100 minutes
 - upload time = 6.40pm - 7.00pm = 20 minutes
- There are 6 questions in the test.
- The start time is same for everyone. So, if you want to use the full 100 minutes of writing time and 20 minutes of upload time, please start right at 5.00pm (Toronto local time)
- Students with accommodations will find their end times on the crowdmark page.
- You are allowed to use lecture notes (the codes, lecture slides, annotated slides or any other files that have been uploaded to Quercus), course lecture recordings, textbooks, R help files.
- Use your best judgment when it comes to the questions during the assessment.

Writing and submitting:

- Write your answers in an R markdown file. Make sure to put page breaks between your answers to the questions. Create an html and save/print it as a pdf using any browser.
- If you know how to create pdf directly from R studio, you can go ahead and do that instead of creating an html and then converting it.
- Make sure to provide the code and the output where ever possible. **Never print large dataset/R object.**
- Leave the last 20 minutes for submitting your answers to crowdmark.
- **Email submissions are not allowed.**
- Students are responsible for making sure that they have uploaded correct and legible answers in correct order in the correct space provided.
- Grading will be done based on the submitted answers only. There is no point of emailing us saying “I accidentally uploaded the wrong file” or “I have uploaded my answer to question 1 in the space provided for question 2”.
- Graders will not look for your answer to one question in the space provided for another question. So please be careful while uploading.
- A late penalty of 2% per minute will be applied after 7:00pm (irrespective of the cause of the delay).
- Submissions made after 7.05pm will automatically get a zero in the test.
- Note: Pressing the submit button at 7:00pm does not mean you have submitted at 7:00pm. Your submission must be complete (including the upload time and any lag that may appear in the system) by 7:00pm.

Academic Integrity:

- Do not share assessment questions or solutions with anyone, at any point of time (even after the test), in any form. It will be considered as an Academic Offence and strict measures will be taken according to the Academic Integrity policies.
- Using any other resources beyond our course is considered an unauthorized aid. This includes but is not limited to:
 - Posting on other discussion board for help (e.g., Slack, Discord, Facebook, etc.)
 - Working with a classmate (this is an individual and independent assessment!).
 - Receiving assistance, tips, hints, or otherwise from someone else.
 - Submitting any work that is not your own.
 - Searching for answers on other platforms (e.g. TAs, Google, Chegg etc.)
- Do not contact anyone (other than the invigilators) during the test time.

Data Sets

- There are two csv files available in the following folder.

https://utoronto-my.sharepoint.com/:f:/g/personal/shahriar_shams_utoronto_ca/EuflxeGppP9CvzfZcU7zufwBhMdVuQ6itUNZPVUEu4PEYg?e=u02aR1

- One of the data set is “Highrise Inspections Data.csv”. This gives properties where violations have been found by an Inspector which are required to be fixed for compliance at the time with the Ontario Fire Code, It also gives properties with no observable violations.
- Another dataset is “Victims of Crime.csv”. This gives a comprehensive overview of police related statistics including reported crimes, victims of crime. Each row gives us total number of victims by each year, assault type, and victim profile (age, sex etc.)
- Download these two data files [use `read.csv()` only, do not use other available similar functions] and upload them to your R studio cloud session on jupyter.utoronto.ca
- Let’s load the “Highrise Inspections Data.csv” as “highrise” and “Victims of Crime.csv” as “crime” in R studio.

```
highrise = read.csv(file="C:/Users/shahr/OneDrive - University of Toronto/STAA57_Teaching/2023_Winter/Midterm1/highrise.csv")
crime = read.csv(file="C:/Users/shahr/OneDrive - University of Toronto/STAA57_Teaching/2023_Winter/Midterm1/victims_of_crime.csv")
```

Question-1 [4 points]

Do these following tasks only for the “highrise” dataset.

- Check the variable names and types. Do they look ok? If you see any variable type that does not match with your expectation then mention it in your answer.
- The column “VIOLATIONS_ITEM_NUMBER” gives you the violation codes. Create a new variable in the dataset named “Violation” which takes value 0 if the “VIOLATIONS_ITEM_NUMBER” is 0, and otherwise takes 1.
- Create a new variable in the dataset that gives you the street names(without the house numbers) based on the “propertyAddress” column.

Question 1 Solution

```
library(tidyverse)

glimpse(highrise)

## Rows: 77,759
## Columns: 10
## $ X_id                <int> 3530512, 3530513, 3530514, 3530515, 3530516, 35~
## $ propertyAddress     <chr> "40 Stevenson Rd", "787 Jane St", "2 Glamorgan ~
## $ propertyType        <chr> "High Rise", "High Rise", "High Rise", "High Ri~
## $ propertyWard        <int> 1, 5, 21, 22, 20, 3, 13, 13, 13, 13, 7, 7, 7, 3~
## $ InspectionThread    <chr> "_643dcACA293CD9B", "_644fcB35B649041", "_645bb~
## $ X                  <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ INSPECTIONS_OPENDATE <chr> "2017-06-05", "2017-02-03", "2017-01-10", "2017~
## $ INSPECTIONS_CLOSEDDATE <chr> "2017-06-05", "2017-02-06", "2017-02-08", "2017~
## $ VIOLATIONS_ITEM_NUMBER <int> 0, 0, 0, 0, 0, 0, 1, 2, 3, 4, 0, 1, 2, 0, 0, 0, ~
## $ Violation_Yes_No    <chr> "No", "No", "No", "No", "No", "No", "Yes", "Yes~
```

INSPECTIONS_OPENDATE and INSPECTIONS_CLOSEDDATE are both dates but were loaded as character variables.

```
highrise = highrise %>% mutate(Violation = as.numeric(VIOLATIONS_ITEM_NUMBER > 0),
                               StreetName = str_remove_all(propertyAddress, "\\d+"))
glimpse(highrise)

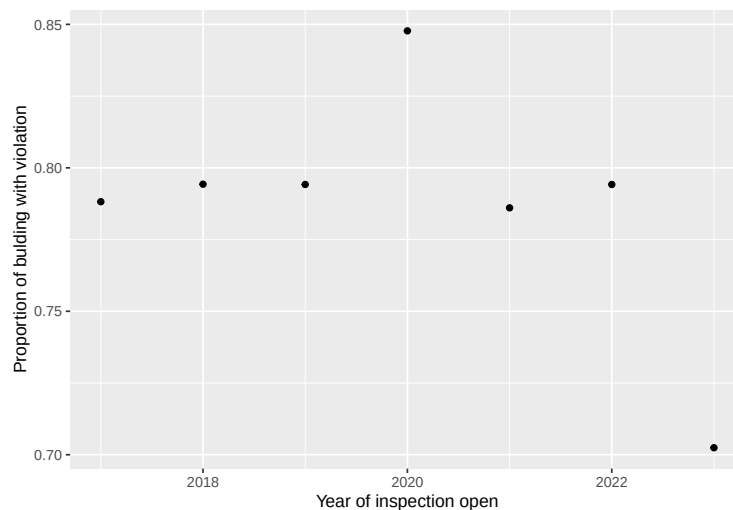
## Rows: 77,759
## Columns: 12
## $ X_id                <int> 3530512, 3530513, 3530514, 3530515, 3530516, 35~
## $ propertyAddress     <chr> "40 Stevenson Rd", "787 Jane St", "2 Glamorgan ~
## $ propertyType        <chr> "High Rise", "High Rise", "High Rise", "High Ri~
## $ propertyWard        <int> 1, 5, 21, 22, 20, 3, 13, 13, 13, 13, 7, 7, 7, 3~
## $ InspectionThread    <chr> "_643dcACA293CD9B", "_644fcB35B649041", "_645bb~
## $ X                  <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
## $ INSPECTIONS_OPENDATE <chr> "2017-06-05", "2017-02-03", "2017-01-10", "2017~
## $ INSPECTIONS_CLOSEDDATE <chr> "2017-06-05", "2017-02-06", "2017-02-08", "2017~
## $ VIOLATIONS_ITEM_NUMBER <int> 0, 0, 0, 0, 0, 0, 1, 2, 3, 4, 0, 1, 2, 0, 0, 0, ~
## $ Violation_Yes_No    <chr> "No", "No", "No", "No", "No", "No", "Yes", "Yes~
## $ Violation           <dbl> 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 0, 1, 1, 0, 0, 0, ~
## $ StreetName          <chr> " Stevenson Rd", " Jane St", " Glamorgan Ave", ~
```

Question-2 [4 points]

Do these following tasks using the “highrise” dataset.

Using the “Violation_Yes_No” column do these following two tasks. [It gives “Yes” if the building has violated a fire code]

- Create a summary table that shows the **proportion** of buildings that violated a code by the **YEAR** of the “INSPECTIONS_OPENDATE”. [hint: you will have to create a year column first]
- Using your summary table from part(i), recreate the following graph



Question 2 Solution

```
library(lubridate)

highrise = highrise %>% mutate(YEAR_OPEN = year(as.Date(INSPECTIONS_OPENDATE)))

high2 = highrise %>% group_by(YEAR_OPEN) %>%
  summarise(prop_violation = mean(Violation_Yes_No == "Yes"))

ggplot(high2, aes(x = YEAR_OPEN, y = prop_violation)) +
  geom_point() +
  labs(x = "Year of inspection open",
       y = "Proportion of bulding with violation")
```

Question-3 [4 points]

Take a random sample of 300 observations from the “highrise” dataset and based on the sample, do these following two tasks.

- i. Construct a 90% confidence interval for the proportion of buildings violating a code.
- ii. Construct a 90% Bootstrap confidence interval for the proportion of buildings violating a code.

Question 3 Solution

```
high3 = highrise %>% sample_n(300)

prop.test(x = sum(high3$Violation_Yes_No == "Yes"),
          n = length(high3$Violation_Yes_No), conf.level = 0.9)

##
## 1-sample proportions test with continuity correction
##
## data:  sum(high3$Violation_Yes_No == "Yes") out of length(high3$Violation_Yes_No), null probability 0.5
## X-squared = 121.6, df = 1, p-value < 2.2e-16
## alternative hypothesis: true p is not equal to 0.5
## 90 percent confidence interval:
##  0.7789254 0.8550951
## sample estimates:
##      p
## 0.82

boot_function=function(){

  boot_data = high3 %>% sample_n(nrow(high3), replace = T)

  boot_prop = mean(boot_data$Violation_Yes_No == "Yes")

  return(boot_prop)

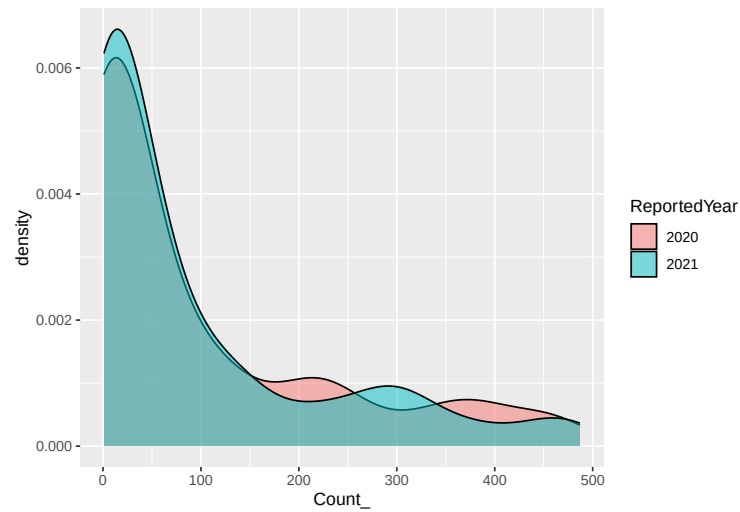
}

quantile(replicate(1000,boot_function()), c(0.05,0.95))

##          5%          95%
## 0.7833333 0.8566667
```

Question-4 [4 points]

Using the “crime” dataset, reproduce the following graph.



Question 4 Solution

```
crime2 = crime %>% filter(ReportedYear %in% c(2020,2021) & Count_<=500)

ggplot(crime2, aes( x=Count_)) +
  geom_density(aes(fill=factor(ReportedYear)),alpha=0.5)+
  labs(fill="ReportedYear")
```

Question-5 [4 points]

Answer this question using the “crime” dataset.

By creating an appropriate summary table, find out the **most frequent** victim description(AgeCohort and Sex) for each year between 2020 and 2021.

[*hint*: most frequent means the category with the largest number of **total count**]

Question 5 Solution

```
crime3 = crime %>%
  filter(ReportedYear >=2020) %>%
  group_by(ReportedYear ,Sex, AgeCohort) %>%
  summarise(total = sum(Count_)) %>%
  arrange(ReportedYear, desc(total)) %>%
  filter(row_number()==1)
crime3
```

```
## # A tibble: 6 x 4
## # Groups:   ReportedYear, Sex [6]
##   ReportedYear Sex   AgeCohort total
##           <int> <chr>   <chr>      <int>
## 1         2020 F     '25-34'   3326
## 2         2020 M     '25-34'   2831
## 3         2020 U     '25-34'     9
## 4         2021 F     '25-34'   3521
## 5         2021 M     '25-34'   2907
## 6         2021 U     '25-34'     7
```

In Both 2020 and 2021, Female with age between 25 and 34 was the most frequent victim category.

Question-6 [5 points]

By using the “crime” dataset, reproduce the following output

ReportedYearSex	Age'<12'	Age'12-17'	Age'18-24'	Age'25-34'	Age'35-44'	Age'45-54'	Age'55-64'	Age'65+'
2018 F	606	1332	2748	3521	2353	1626	825	616
2018 M	597	1369	2444	3111	2250	2099	1352	637
2018 U	1	NA	2	5	3	3	NA	NA
2019 F	616	1407	2526	3504	2431	1580	929	655
2019 M	694	1526	2501	3195	2216	1891	1398	670
2019 U	3	3	5	7	2	1	NA	NA
2020 F	466	913	2171	3326	2258	1437	806	575
2020 M	451	739	1805	2831	1897	1752	1340	670
2020 U	1	2	3	9	2	6	3	1

where, the numbers in the table represents the total number of counts for each year, sex and age category.

Question 6 Solution

```
crime4 = crime %>%
  filter(ReportedYear %in% c(2018,2019,2020)) %>%
  group_by(ReportedYear ,Sex, AgeCohort) %>%
  summarize(total = sum(Count_))
```

```
head(crime4)
```

```
## # A tibble: 6 x 4
## # Groups:   ReportedYear, Sex [1]
##   ReportedYear Sex   AgeCohort total
##         <int> <chr> <chr>      <int>
## 1         2018 F     '<12'      606
## 2         2018 F     '12-17'    1332
## 3         2018 F     '18-24'    2748
## 4         2018 F     '25-34'    3521
## 5         2018 F     '35-44'    2353
## 6         2018 F     '45-54'    1626
```

```
crime5=crime4 %>% pivot_wider(id_cols= c(ReportedYear, Sex),
                              names_from = AgeCohort,
                              names_prefix = "Age",
                              values_from = total)
```

```
library(knitr)
kable(crime5)
```